



ISTN7MP MOBILE AND PERSONAL INFORMATION SYSTEMS HONOURS MODULE – SEMESTER 1 - 2026

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MOBILE AND PERSONAL INFORMATION SYSTEMS

- PART I : MOBILE COMPUTING
- PART II : MOBILE AND PERSONAL INFORMATION SYSTEMS
- PART III : ANDROID MOBILE APP DEVELOPMENT

PART III

MOBILE OS TRENDS & ARCHITECTURE
AND
ANDROID APPLICATION DEVELOPMENT

MOBILE APPLICATION DEVELOPMENT

ANDROID APP DEVELOPMENT

APP DEMO – 5

CARS SEARCH FINANCING APP

EXPLORE! PERSISTENT DATA

CARS SEARCH FINANCING APP

- THIS APPLICATION IS DESIGNED TO CONTAIN TWO ACTIVITIES
- UKZN CAR LOAN SCHEME PROVIDES LOWER INTEREST RATE FOR THE EMPLOYEES FOR A DURATION OF EITHER 3, 4 OR 5 YEARS
- CARS SEARCH FINANCING IS A MOBILE APPLICATION THAT CALCULATES THE MONTHLY PAYMENT FOR THE EMPLOYEES OPTING FOR THE CAR LOAN EITHER FOR 3, 4 OR 5 YEARS
- THIS APPLICATION DEMONSTRATES THE USE OF PERSISTENT DATA – STORING DATA RECEIVED AS INPUT IN THE FIRST ACTIVITY AND MAKE USE OF THE STORED DATA IN THE SECOND ACTIVITY

CARS SEARCH FINANCING APP (CONTD...)

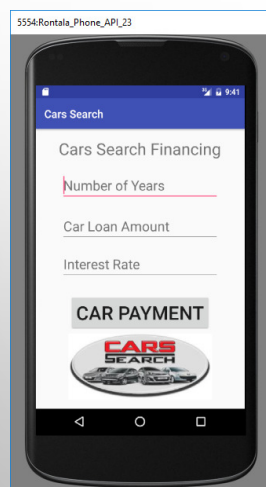
- THE FIRST ACTIVITY CAPTURES INPUT
- THE DATA PROVIDED AS INPUT IS STORED AS PERSISTENT DATA
- ON CLICK OF THE BUTTON TO CALCULATE MONTHLY INSTALLMENT THE EXECUTION CONTROL NAVIGATES TO THE SECOND ACTIVITY
- THE SECOND ACTIVITY RETRIEVES THE STORED PERSISTENT DATA
- CALCULATES THE MONTHLY INSTALLMENT
- DISPLAYS MONTHLY INSTALLMENT ALONG WITH AN IMAGE THAT INDICATES WHETHER THE INSTALLMENT IS FOR 3, 4 OR 5 YEARS. DIFFERENT IMAGES ARE USED FOR DIFFERENT YEARS

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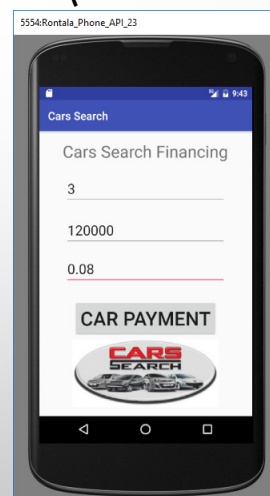
3/29/2026

7

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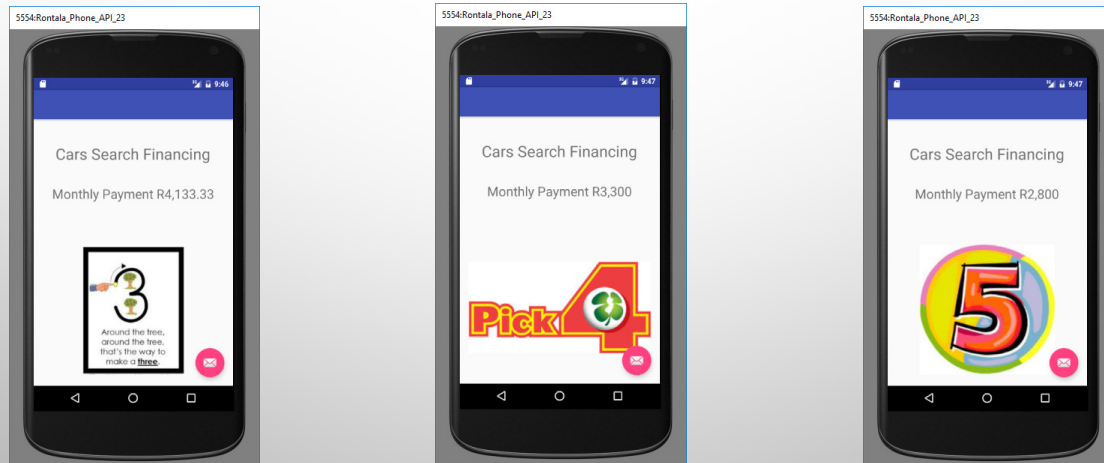
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8

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CARS SEARCH FINANCING APP – PERSISTENT DATA

- PERSISTENT DATA STORES VALUES PERMANENTLY BY PLACING THE INFORMATION IN A FILE
- ANDROID APPLICATIONS CAN SAVE DATA ON THE DEVICE'S DRIVE OR OTHER STORAGE MEDIA SUCH AS A MEMORY CARD OR CLOUD SERVICE SO THAT THE DATA BE RETRIEVED LATER WITHIN THE APP OR AFTER THE TERMINATION OF THE PROGRAM
- ANDROID PROVIDES SEVERAL OPTIONS FOR YOU TO SAVE PERSISTENT APPLICATION DATA

CARS SEARCH FINANCING APP – PERSISTENT DATA (CONTD...)

- PERSISTENT DATA CAN BE SAVED WITHIN A VARIABLE OR WITHIN A DATABASE, BASED ON THE SPECIFIC NEED OF THE APP.
- FIVE DIFFERENT WAYS OF STORING PERSISTENT DATA IN ANDROID APPS
 - SHARED PREFERENCES – STORES PRIVATE DATA IN KEY-VALUE PAIRS
 - INTERNAL STORAGE – STORES PRIVATE DATA IN THE MEMORY OF THE DEVICE
 - EXTERNAL STORAGE – STORES DATA, WHICH CAN BE AVAILABLE TO OTHER APPS ON SHARED EXTERNAL STORAGE
 - SQLITE DATABASE – STORES STRUCTURED DATA IN A PRIVATE DATABASE
 - NETWORK CONNECTION – STORES DATA ON A WEB SERVER

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11

CARS SEARCH FINANCING APP – PERSISTENT DATA – USING SHARED PREFERENCES

- THE SHARED PREFERENCES CLASS PROVIDES ONE OF THE EASIEST WAYS TO SAVE AND LOAD PRIMITIVE DATA
- FOR EXAMPLE,
 - USER'S NAME OR SETTINGS SUCH AS THE FONT SIZE THE USER PREFERS
 - USER'S PREFERENCES SUCH AS WALLPAPER A USER HAS CHOSEN OR INDIVIDUAL VALUES ENTERED BY THE USER IN AN EDITTEXT CONTROL
- SHARED PREFERENCES CAN BE USED TO SAVE ANY PRIMITIVE DATA: BOOLEAN, FLOATS, INTS, LONGS, AND STRINGS.
- SHARED PREFERENCES ARE THE BEST WHEN YOUR APP NEEDS TO SAVE SMALL CHUNKS OF DATA SUCH AS NAME-VALUE PAIR

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12

CARS SEARCH FINANCING APP – PERSISTENT DATA – USING SHARED PREFERENCES (CONTD...)

- THE TWO-PART PAIR SPECIFIES A NAME FOR THE DATA THAT NEED TO BE SAVED AND THE ACTUAL VALUE
- THE PAIR IS SAVED IN AN XML FILE THAT CAN BE RETRIEVED LATER IN THE APP OR AFTER THE APP CLOSES
- STEPS TO SAVE DATA IN A SHARED PREFERENCES FILE
 - OBTAIN AN INSTANCE OF THE SHARED PREFERENCE FILE
 - CREATE A SHAREDREFERENCES.EDITOR OBJECT
 - ASSIGN VALUES INTO SHAREDREFERENCES OBJECTS USING THE PUTSTRING() METHOD
 - SAVE THE VALUES TO THE PREFERENCES FILE USING THE APPLY () METHOD (IN PREVIOUS VERSIONS ITS CALLED COMMIT() METHOD)
- IN THE CARS SEARCH FINANCING APP, NUMBER OF YEARS (INT), LOAN AMOUNT (FLOAT) AND THE INTEREST RATE (FLOAT) CAPTURED AS INPUT IN THE FIRST ACTIVITY IS SAVED AND RETRIEVED IN THE SECOND ACTIVITY

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13

CARS SEARCH FINANCING APP – KNOWLEDGE REQUIRED

- ADD STRINGS TO THE STRING TABLE
- ADD IMAGES TO THE DRAWABLE FOLDER
- DESIGN TWO XML LAYOUTS FOR THE FIRST AND SECOND ACTIVITY
- INSTANTIATE THE XML CONTROLS IN THE FIRST ACTIVITY
- ESTABLISH A SHARED PREFERENCES OBJECT TO STORE THE DATA ENTERED
- WRITE DATA TO THE SHARED PREFERENCES OBJECT
- LAUNCH A SECOND ACTIVITY
- INITIALIZE THE XML CONTROLS ON THE SECOND ACTIVITY'
- RETRIEVE THE DATA FROM THE SHARED PREFERENCES OBJECT
- CALCULATE THE MONTHLY PAYMENT
- DISPLAY THE APPROPRIATE IMAGE INDICATING THE NUMBER OF LOAN YEARS USING ONE IMAGE VIEW WITH IF STRUCTURE
- DISPLAY THE MONTHLY PAYMENT ON THE SECOND ACTIVITY

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14

CARS SEARCH FINANCING APP – WHAT’S NEW IN THIS APP

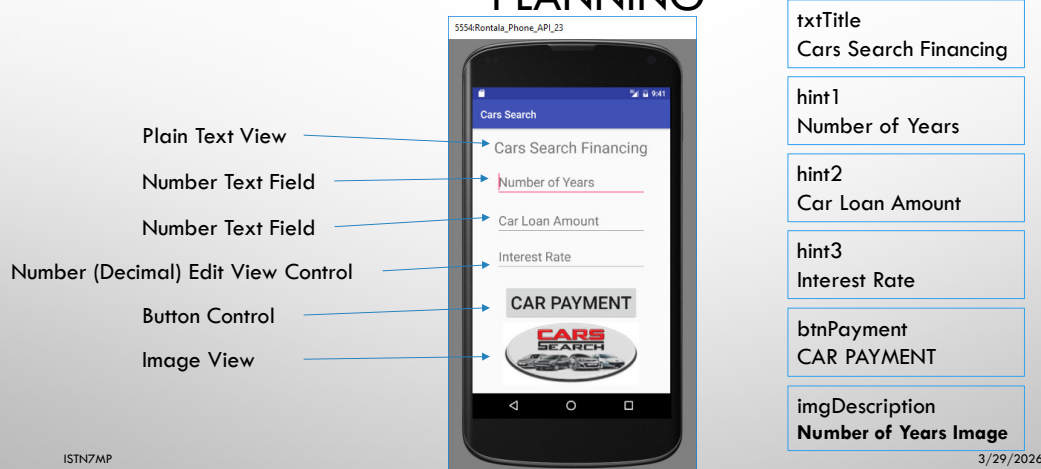
- ADD STRINGS TO THE STRING TABLE
- ADD IMAGES TO THE DRAWABLE FOLDER
- DESIGN TWO XML LAYOUTS FOR THE FIRST AND SECOND ACTIVITY
- INSTANTIATE THE XML CONTROLS IN THE FIRST ACTIVITY
- **ESTABLISH A SHARED PREFERENCES OBJECT TO STORE THE DATA ENTERED**
- **WRITE DATA TO THE SHARED PREFERENCES OBJECT**
- LAUNCH A SECOND ACTIVITY
- INITIALIZE THE XML CONTROLS ON THE SECOND ACTIVITY
- **RETRIEVE THE DATA FROM THE SHARED PREFERENCES OBJECT**
- CALCULATE THE MONTHLY PAYMENT
- **DISPLAY THE APPROPRIATE IMAGE INDICATING THE NUMBER OF LOAN YEARS USING ONE IMAGE VIEW WITH IF STRUCTURE**
- **DISPLAY THE MONTHLY PAYMENT ON THE SECOND ACTIVITY**

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15

CARS SEARCH FINANCING APP – USER INTERFACE PLANNING



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16

STEP 1: ADDING STRINGS

- ADD FOLLOWING STRINGS INTO “STRINGS.XML” FILE IN “RES” FOLDER THROUGH TRANSLATIONS EDITOR

Key	Default Value
txtTitle	Cars Search Financing
hint1	Number of Years
hint2	Car Loan Amount
hint3	Interest Rate
btnPayment	CAR PAYMENT
imgDescription	Number of Years Image
txtPayment	Monthly Payment

STEP 2: ADDING IMAGES

- FOUR IMAGES ARE REQUIRED

ONE WHILE STARTING THE APPLICATION IN THE MAIN ACTIVITY (START.JPG)

TWO, TO INDICATE THREE YEARS OF LOAN DURATION (THREE.JPG)

THREE, TO INDICATE FOUR YEARS OF LOAN DURATION (FOUR.JPG)

FOUR, TO INDICATE FIVE YEARS OF LOAN DURATION (FIVE.JPG)

- COPY ALL FOUR IMAGES FROM THE STORED LOCATION AND PASTE THEM INTO THE DRAWABLE FOLDER

STEP 3 – ADDING WIDGETS & CONTROLS (MAIN ACTIVITY)

- PLAIN TEXT VIEW FOR THE TITLE
- NUMBER CONTROL FROM TEXT FIELDS CATEGORY FOR NUMBER OF YEARS
- NUMBER CONTROL (EDITVIEW) CONTROL FROM TEXT FIELDS CATEGORY FOR LOAN AMOUNT
- NUMBER (DECIMAL) EDIT VIEW CONTROL FROM TEXT FIELDS CATEGORY FOR INTEREST RATE
- BUTTON CONTROL
- IMAGE VIEW CONTROL FOR THE START IMAGE

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19

STEP 4 – XML CODE FOR WIDGETS & CONTROLS (MAIN ACTIVITY – ACTIVITY_MAIN.XML)

<TextView

```

    android:id="@+id/txtTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/txtTitle"

```

```

app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.098" />

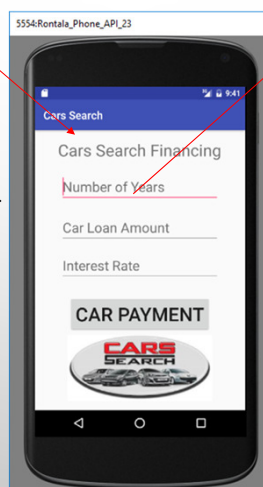
```

<EditText

```

    android:id="@+id/txtYears"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="@string/hint1"
    android:inputType="number"
    android:textSize="24sp"
    android:autofillHints="no"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.511"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.181" />

```



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20

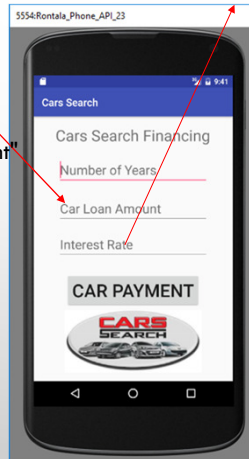
STEP 4 – XML CODE FOR WIDGETS & CONTROLS (MAIN ACTIVITY – ACTIVITY_MAIN.XML) [CONTD...]

<EditText

```
android:id="@+id/txtLoan"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:hint="@string/hint2"
android:inputType="numberDecimal"
android:textSize="24sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.51"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.29" />
```

<EditText

```
android:id="@+id/txtInterest"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:hint="@string/hint3"
android:inputType="numberDecimal"
android:textSize="24sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.509"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.418" />
```



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21

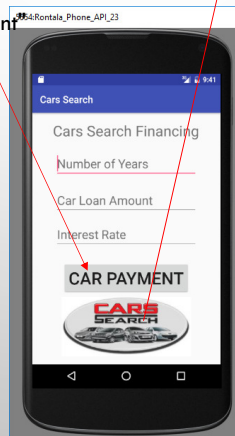
STEP 4 – XML CODE FOR WIDGETS & CONTROLS (MAIN ACTIVITY – ACTIVITY_MAIN.XML) [CONTD...]

<Button

```
android:id="@+id/btnPayment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/btnPayment"
android:textSize="34sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.505"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.59" />
```

<ImageView

```
android:id="@+id/imgStart"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:contentDescription="@string/imgDescription"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.82"
app:srcCompat="@drawable/start" />
```



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22

STEP 5 – CREATING SECOND ACTIVITY

- a) EXPAND JAVA FOLDER WITHIN THE ANDROID PROJECT VIEW
- b) RIGHT CLICK ON THE FIRST FOLDER WITHIN JAVA FOLDER WHERE MAINACTIVITY.JAVA IS AVAILABLE
- c) INSERT A NEW ACTIVITY BY CHOOSING NEW -> ACTIVITY -> EMPTY ACTIVITY AND CAPTURE ALL REQUIRED INFORMATION AND NAME THE ACTIVITY AS PAYMENT

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23

STEP 6 – ADDING WIDGETS & CONTROLS (PAYMENT ACTIVITY)

- PLAIN TEXT VIEW FOR THE TITLE
- ANOTHER PLAIN TEXT VIEW TO DISPLAY THE RESULT
- IMAGE VIEW CONTROL TO DISPLAY ONE OF THE THREE POSSIBLE IMAGES

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24

STEP 7 – XML CODE FOR WIDGETS & CONTROLS (CONTENT_PAYMENT.XML)

```

<TextView
    android:id="@+id/txtTitle2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/txtTitle"
    android:textSize="34sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.458"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.113" />

<ImageView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/imgYears"
    android:contentDescription="@string/imgDescription"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.82" />

<TextView
    android:id="@+id/txtMonthlyPayment"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text=""
    android:textAlignment="center"
    android:textSize="20sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.352" />
  
```

STEP 8 – INSTANTIATING THE XML CONTROLS

- WITHIN MAINACTIVITY.JAVA INCLUDE THE FOLLOWING CODE AFTER SETCONTENTVIEW METHOD TO DEFINE AN INSTANCES FOR CAPTURING INPUT (NUMERIC TEXTVIEW) AND A BUTTON

```

final EditText years = findViewById(R.id.txtYears);
final EditText loan = findViewById(R.id.txtLoan);
final EditText interest = findViewById(R.id.txtInterest);
Button button = findViewById(R.id.btnPayment);
  
```

STEP 8 – INSTANTIATING THE XML CONTROLS (CONTD...)

- WITHIN MAINACTIVITY.JAVA INCLUDE THE FOLLOWING CODE TO EXECUTE ON CLICK OF THE BUTTON

```
button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

    }
});
```

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3/29/2026

27

STEP 8 – INSTANTIATING THE XML CONTROLS (CONTD...)

- WITHIN MAINACTIVITY.JAVA INCLUDE THE FOLLOWING CODE TO DECLARE REQUIRED VARIABLES TO CAPTURE INPUT

```
button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        intYears=Integer.parseInt(years.getText().toString());
        intLoan=Integer.parseInt(loan.getText().toString());
        decInterest=Float.parseFloat(interest.getText().toString());

    }
});
```

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3/29/2026

28

STEP 9 – WRITING PERSISTENT DATA WITH SHARED PREFERENCES

- THE DATA IS SAVED TO AN XML FILE AS A **KEY-VALUE** PAIR
- THE **KEY** IS A STRING SUCH AS “**KEY1**” THAT UNIQUELY IDENTIFIES THE PREFERENCE, AND THE **VALUE** IS THE DATA REPRESENTED AS A STRING, INT, LONG, FLOAT, OR BOOLEAN
- THE FOLLOWING DATA TYPES ARE SUPPORTED BY THE SHAREDREFERENCE CLASS:
 - PUTSTRING() – STORES STRING VALUES
 - PUTINT() – STORES INTEGER VALUES
 - PUTLONG() – STORES LONG VALUES
 - PUTFLOAT() – STORES FLOAT VALUES
 - PUTBOOLEAN() – STORES BOOLEAN VALUES

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29

STEP 9 – WRITING PERSISTENT DATA WITH SHARED PREFERENCES (CONTD...)

- WITHIN MAINACTIVITY.JAVA INCLUDE THE FOLLOWING CODE FOR STORING DATA USING SHARED PREFERENCES

```
button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        intYears = Integer.parseInt(years.getText().toString());
        intLoan = Integer.parseInt(loan.getText().toString());
        declInterest = Float.parseFloat(interest.getText().toString());
        SharedPreferences sharedPref = getSharedPreferences("MyPrefs", MODE_PRIVATE);
        SharedPreferences.Editor editor;
        editor = sharedPref.edit();
        editor.putInt("key1", intYears);
        editor.putInt("key2", intLoan);
        editor.putFloat("key3", declInterest);
        editor.apply();
        startActivity(new Intent(MainActivity.this, Payment.class));
    }
});
```

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3/29/2026

30

STEP 10 – LAUNCHING THE SECOND ACTIVITY

- WITHIN MAINACTIVITY.JAVA INCLUDE THE FOLLOWING CODE TO START THE SECOND ACTIVITY

```
button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        intYears = Integer.parseInt(years.getText().toString());
        intLoan = Integer.parseInt(loan.getText().toString());
        declInterest = Float.parseFloat(interest.getText().toString());
        SharedPreferences sharedPref = getSharedPreferences("MyPrefs", MODE_PRIVATE);
        SharedPreferences.Editor editor;
        editor = sharedPref.edit();
        editor.putInt("key1", intYears);
        editor.putInt("key2", intLoan);
        editor.putFloat("key3", declInterest);
        editor.apply();
        startActivity(new Intent(MainActivity.this, Payment.class));
    }
});
```

ISTN7MP 3/29/2026 31

STEP 11 – INSTANTIATING THE SECOND ACTIVITY CONTROLS

- WITHIN PAYMENT.JAVA INCLUDE THE FOLLOWING CODE

```
TextView monthlyPayment=findViewById(R.id.txtMonthlyPayment);
ImageView image=findViewById(R.id.imgYears);
```

ISTN7MP 3/29/2026 32

STEP 12 – RETRIEVING PREFERENCES

- RETRIEVING DATA IS AS EASY AS SAVING IT
- AN EDITOR IS NOT REQUIRED TO READ THE SAVED DATA
- DATA CAN BE RETRIEVED FROM SHARED PREFERENCE OBJECT BY USING THE APPROPRIATE METHODS TO RETRIEVE A KEY'S VALUE BY NAME:
 - GETSTRING() – RETRIEVES STRING VALUES
 - GETINT() – RETRIEVES INTEGER VALUES
 - GETLONG() – RETRIEVES LONG VALUES
 - GETFLOAT() – RETRIEVES FLOAT VALUES
 - GETBOOLEAN() – RETRIEVES BOOLEAN VALUES

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3/29/2026

33

STEP 12 – RETRIEVING PREFERENCES (CONTD...)

- WITHIN PAYMENT.JAVA INCLUDE THE FOLLOWING CODE TO RETRIEVE STORED DATA USING SHARED PREFERENCES

```
SharedPreferences sharedPref = getSharedPreferences("MyPrefs", MODE_PRIVATE);  
int intYears=sharedPref.getInt("key1",0);  
int intLoan=sharedPref.getInt("key2",0);  
float decInterest=sharedPref.getFloat("key3",0);
```

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3/29/2026

34

STEP 13 – CALCULATING MONTHLY INSTALLMENT

- WITHIN PAYMENT.JAVA INCLUDE THE FOLLOWING CODE FOR NUMBER FORMATTING AFTER THE CODE FOR SHARED PREFERENCES

```
float decMonthlyPayment;
decMonthlyPayment=(intLoan*(1+((decInterest/100)*intYears)))/(12*intYears);
DecimalFormat currency=new DecimalFormat("R###,###.##");
monthlyPayment.setText("Monthly Payment "+currency.format(decMonthlyPayment));
```

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3/29/2026

35

STEP 14 – CODING AN IMAGE VIEW CONTROL

- WITHIN PAYMENT.JAVA INCLUDE THE FOLLOWING CODE TO SELECT THE RIGHT IMAGE TO BE DISPLAYED

```
if(intYears==3){
    image.setImageResource(R.drawable.three);
} else if (intYears==4) {
    image.setImageResource(R.drawable.four);
} else if (intYears==5) {
    image.setImageResource(R.drawable.five);
} else {
    monthlyPayment.setText("Enter 3, 4, or 5 Years");
}
```

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3/29/2026

36

STEP 15 – RUNNING & TESTING THE APPLICATION

- SAVE ALL THE WORK DONE
- RUN THE APPLICATION

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3/29/2026

37

ANDROID PRACTICE ACTIVITY 5

1. PRACTICE THE SAME APP DEMONSTRATED DURING THE LECTURE SESSION
2. MODIFY THE CARS SEARCH APP BY REPLACING SHARED PREFERENCES WITH SQLITE. IMPROVE THE APP BY CAPTURING CUSTOMER PERSONAL DETAILS, CAR DETAILS, AND COST DETAILS.
 - a) THE FIRST ACTIVITY WILL CAPTURE CUSTOMER'S PERSONAL DETAILS
 - b) THE SECOND ACTIVITY WILL CAPTURE THE CAR DETAILS LIKE MAKE, MODEL, COLOR, TYPE, ETC.
 - c) THE CURRENT FIRST ACTIVITY WILL BE THE THIRD ACTIVITY IN THE MODIFIED CARS SEARCH APP
 - d) THE CURRENT SECOND ACTIVITY WILL BE THE FOURTH ACTIVITY IN THE MODIFIED CARS SEARCH APP

NOTE: WITH THE MINIMAL INFORMATION PROVIDED ABOVE, APPLY CREATIVITY AND INNOVATIVE IDEAS IN DEVELOPING THE ABOVE APP

TIP: ONE CUSTOMER MAY PURCHASE ONE OR MORE CARS

THANK YOU

END OF
MOBILE APP DEVELOPMENT: APP DEMO 5

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3/29/2026

39